

2 Mixtures and separations

2.1

Elements and compounds

LEARNING OUTCOMES

At the end of this topic you should be able to:

- explain the terms element and compound
- explain that elements and compounds are pure substances
- explain the difference between physical and chemical properties.

EXAM TIP

When you describe an element, make sure that it is clear that *all* the atoms in the element are of one type only. An answer such as 'it contains only one atom' is not correct.

DID YOU KNOW?

Some forms of chemical elements are made artificially and can only exist for a few seconds. For example, rutherfordium-260 can only exist for less than a second.

Elements

An **element** is a substance made up of only one type of atom. Elements cannot be broken down into anything simpler by chemical reactions.

Some examples of elements are shown in Figure 2.1.1. The structures look different but each one of them only has one type of atom. Chlorine has only chlorine atoms, carbon has only carbon atoms and so on.

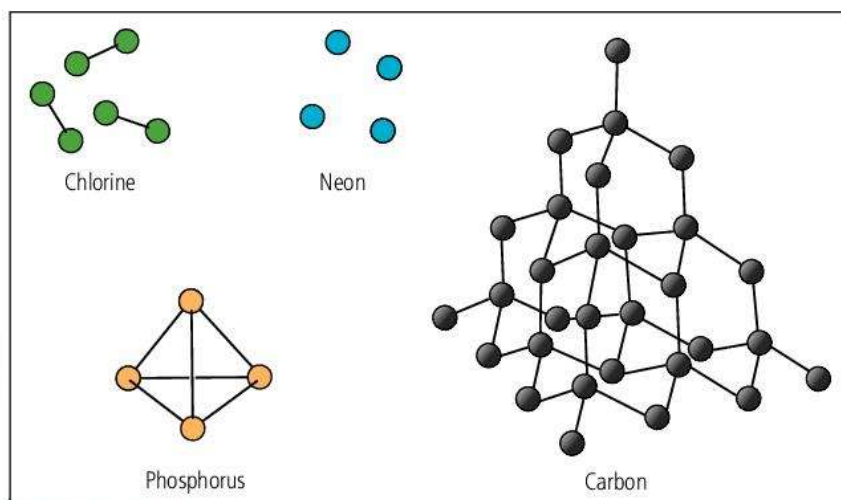


Figure 2.1.1 Each of these four elements is only made up of one type of atom.

Chemical bonds

In Figure 2.1.1, the atoms of chlorine, phosphorus and carbon are joined by chemical bonds.

- A chemical bond is shown by a line joining the atoms.
- A chemical bond shows that there are strong forces holding the atoms together.
- The bonds shown here are called covalent bonds (see 5.2).

Compounds

A **compound** is a substance made up of two or more different atoms (or ions) joined together by bonds.

Compounds always have a fixed amount of each element in them. For example, a water molecule always has two hydrogen atoms and one oxygen atom. In sodium chloride (salt) the ratio of sodium ions to chloride ions is always 1 : 1 (see Figure 2.1.2).

There are two main groups of compounds:

- molecular (or giant molecular), where atoms are bonded together (see 5.2)
- ionic, where positive and negative ions form 'ionic bonds' (see 5.1).

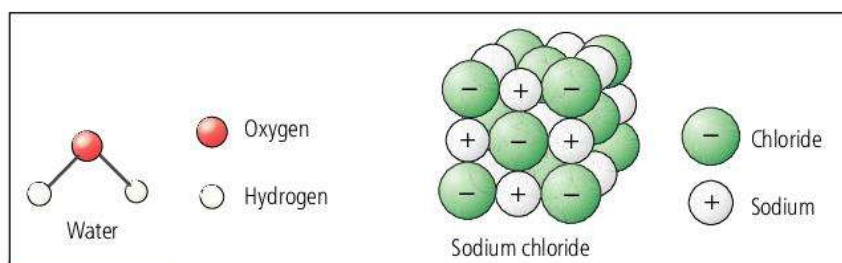


Figure 2.1.2 Water is a molecular compound of hydrogen and oxygen. Sodium chloride is an ionic compound of sodium and chloride ions.

Physical and chemical properties

Physical properties are ones that do not generally depend on the amount of substance present. Examples are:

- melting and boiling points
- density (mass of substance divided by volume of substance)
- strength
- hardness
- electrical and thermal conductivity.

Chemical properties describe how elements and compounds react with other substances. For example:

- Sodium reacts with chlorine to form sodium chloride.
- Sodium reacts with water to form sodium hydroxide and hydrogen.

Reacting elements

Compounds often have very different properties from the elements they are made from. We can see this by comparing the properties of sodium, chlorine and sodium chloride. Sodium chloride is the compound formed by the chemical reaction when sodium burns in chlorine gas.

Table 2.1.1 Some physical properties of sodium, chlorine and sodium chloride

	Sodium	Chlorine	Sodium chloride
State	Solid	Gas	Solid
Colour	Silvery	Green	White
Boiling point (°C)	883	−35	801
Density (g cm^{−3}) at room temperature	0.97	0.00296	2.17

Table 2.1.2 Some chemical properties of sodium, chlorine and sodium chloride

	Sodium	Chlorine	Sodium chloride
Reaction with oxygen	Burns to form a white powder	Does not react	Does not react
Reaction with water	Reacts to form an alkaline solution	Reacts to form an acidic solution	Does not react – just dissolves

EXAM TIP

Remember that there are two parts to the definition of a compound:

- There is more than one type of atom.
- The atoms are bonded (joined) together.

KEY POINTS

- 1 An element is a substance made up of only one type of atom.
- 2 A compound is a substance made up of two or more different atoms bonded (joined) together.
- 3 Compounds have physical and chemical properties that are different from those of the elements they are made from.

LEARNING OUTCOMES

At the end of this topic you should be able to:

- explain the terms fixed composition and variable composition
- explain the differences between compounds and mixtures
- use differences in boiling points to describe the differences between pure and impure substances
- explain that mixtures can be separated by filtration and other physical means.

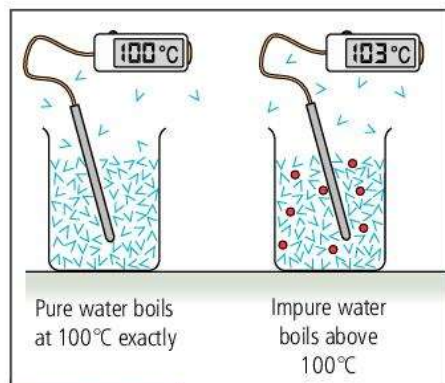


Figure 2.2.2

DID YOU KNOW?

In countries that have cold winters, salt is spread on the road to try to stop ice forming and causing accidents. The melting point of water is lowered by adding salt. This means that water only turns to ice at temperatures some way below 0°C.

Pure substance or mixture?

Elements and compounds are examples of pure substances. If a substance is pure, we cannot separate it into any other parts (components) by physical means. Pure water has only one component in it – water molecules. Pure sulfur has only one component in it – sulfur molecules. The composition of a compound is fixed (see 2.1).

A **mixture** is an impure substance that contains two or more different components. It consists of two or more elements or compounds which are not chemically bonded together (see Figure 2.2.1). A solution of sodium chloride (salt) in water, for example, is a mixture because it contains two components: salt and water.

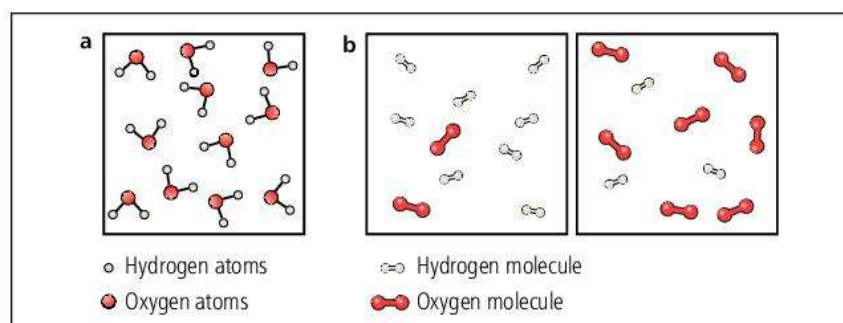


Figure 2.2.1

a Water is a pure compound. It contains only water molecules in which there are two hydrogen atoms for every oxygen atom.

b A mixture of hydrogen and oxygen can have varying amounts of hydrogen and oxygen in it.

Pure or impure: comparing boiling points

Pure substances have sharp melting points and boiling points. The melting point of water is exactly 0°C and the boiling point is exactly 100°C at atmospheric pressure.

If a substance is impure, the impurities will affect its melting point and boiling point.

- The boiling point of a liquid is increased by adding other substances. Making water impure by dissolving a lot of sodium chloride in it raises the boiling point from 100°C to about 103°C (Figure 2.2.2). The more salt that is dissolved, the greater the rise in the boiling point.
- The melting point of a substance is decreased by adding impurities. Adding salt to water makes it freeze at temperatures below 0°C.

Mixtures can be separated by physical means

The components of a mixture can be separated by physical means such as filtration or distillation. For example, a mixture of sand and salt can be separated by:

- 1 dissolving the salt in water – the sand does not dissolve
- 2 filtering the mixture – the sand remains on the filter paper and the salt solution goes through
- 3 the water is evaporated from the salt solution to leave salt (Figure 2.2.3).

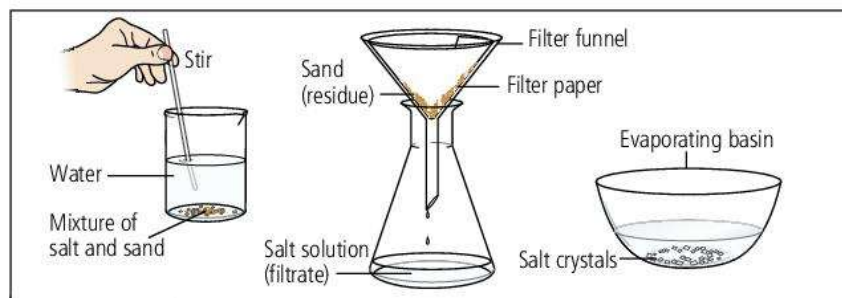


Figure 2.2.3 Separating salt from sand

Comparing compounds and mixtures

Table 2.2.1 Differences between a compound and a mixture

Compound (pure substance)	Mixture
Composition is fixed	Composition is variable
Cannot be separated by physical means	Can be separated by physical means
Physical properties are different from the elements from which they are made.	Physical properties, e.g. colour and density, are the average of the substances in the mixture.

Comparing a mixture of iron and sulfur with the compound iron sulfide

Iron is a silvery metal and sulfur is a yellow non-metal. Iron sulfide is a compound of iron and sulfur. It is made when iron and sulfur are heated together. It contains sulfur atoms bonded to iron atoms. Some of the differences between the mixture (impure substance) and the compound (pure substance) are shown in Table 2.2.2.

Table 2.2.2 Differences between iron sulfide and a mixture of iron and sulfur

Iron and sulfur (mixture)	Iron sulfide (compound)
The colour is a mixture of tiny yellow and silver spots.	The colour is brown-black.
The iron can be separated from the sulfur using a magnet.	The iron cannot be separated from the sulfur using a magnet.
The sulfur can be dissolved in an organic solvent leaving the iron as a solid.	Iron sulfide does not dissolve in organic solvents.
When the mixture is formed, no heat is given off.	When the compound is formed, heat is given off.
Hydrogen gas is formed (by reaction with iron) when an acid is added. The gas does not smell.	Hydrogen sulfide gas is formed when an acid is added. The gas smells of bad eggs.

EXAM TIP

Remember that although compounds have several different types of atoms in them, a pure compound always has the same ratio of atoms of each type. A mixture can have varying numbers of each type of atom because you can change the ratio of the substances you add together.



Figure 2.2.4 This waste pond is a mixture of water and chemical waste from the extraction of aluminium.

KEY POINTS

- 1 Pure substances contain only one component.
- 2 Mixtures contain two or more elements or compounds but they are not chemically bonded together.
- 3 The boiling point of a pure compound is increased if it made impure by adding another substance.
- 4 The components of a mixture can be separated by physical means, but it is not possible to separate the different atoms in a compound by physical means.
- 5 The composition of a mixture is variable but the composition of a compound is fixed.